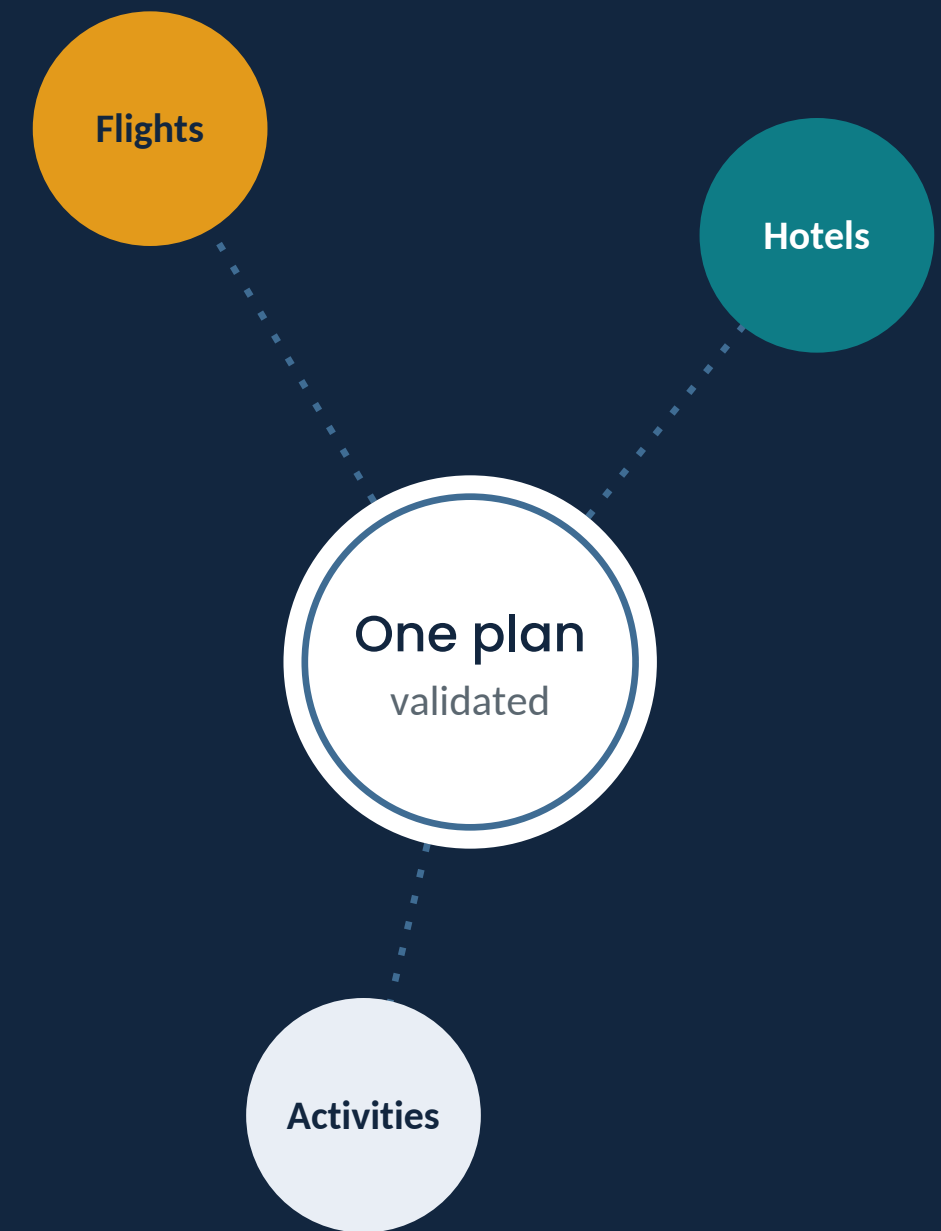


Trip Genie

A Multi-Agent AI Travel Planner

Product presentation | Agentic AI portfolio project
Built on n8n with Google Gemini, SerpApi and Tavily

Animesh Deb | September 2026



The problem: planning a trip is a multi-tab chore

Fragmented

Flights, hotels and activities live in different tools. Travellers re-type the same cities and dates again and again.

Unreconciled

Nobody checks that hotel nights, flight dates and activity days line up. The traveller reconciles it by hand.

Untrustworthy chatbots

General chatbots cannot search live inventory and fill gaps with guesses: invented cities, dates and prices.

Painful refinement

"Two more hotels" means starting over or seeing options that were already rejected.

Week-0 baseline to capture:

Time a manual plan (flights + hotel + activities) for 3-5 real trips, then compare against Trip Genie on the same trips.

Who it is for and the job to be done

Working personas: hypotheses to validate with 5-8 traveller interviews

Time-poor professional

Knows the city and date; wants a credible shortlist fast.

WHAT MATTERS MOST

Speed, credible prices, no rework

Group or family organiser

Coordinates a multi-day holiday for several people.

WHAT MATTERS MOST

Consistent dates, hotel class, easy refinement

Exploratory leisure traveller

Has a destination in mind, not the logistics.

WHAT MATTERS MOST

Variety, local flavour, inspiration

Job to be done

When I have a city and a date, show me a realistic plan I can trust, and let me refine it in conversation without starting over.

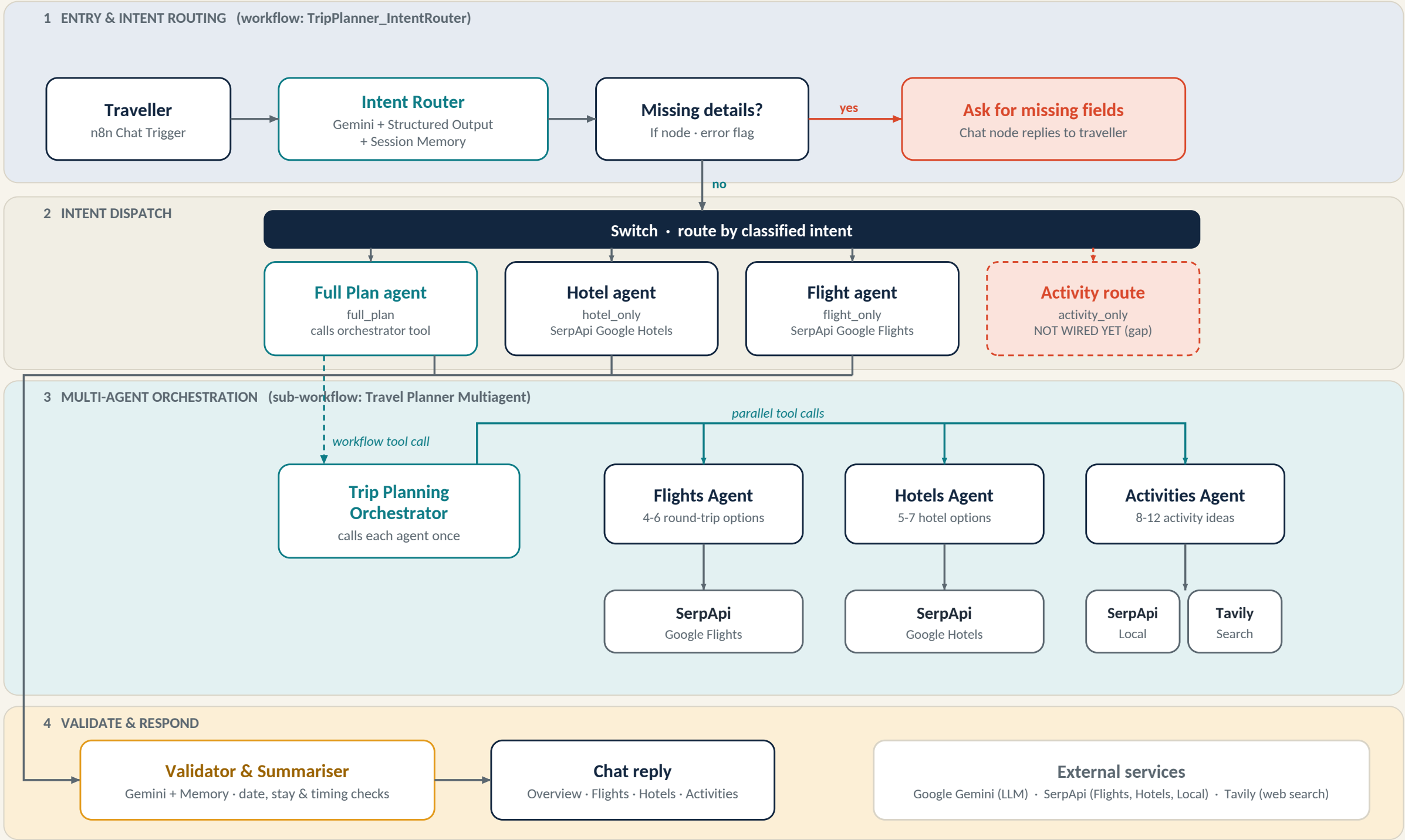
The solution: an agentic workflow with a conscience



Design principles

- Ask, don't guess: a blank field is correct, a guessed field is wrong.
- Route small, reuse big: single-domain asks skip the full orchestration.
- Ground every option in tool output; agents summarise, they do not invent.
- Keep the plumbing invisible: travellers never see workflow or system details.

Architecture: four layers, three n8n workflows



Smart intent routing: ask, don't guess

Four intents, one router

full_plan	"Mumbai to Goa, 15th December, 4 days"	Built
hotel_only	"I need 2 more hotel options"	Built
flight_only	"Any direct flights available?"	Partial
activity_only	"What else is there to do?"	Gap

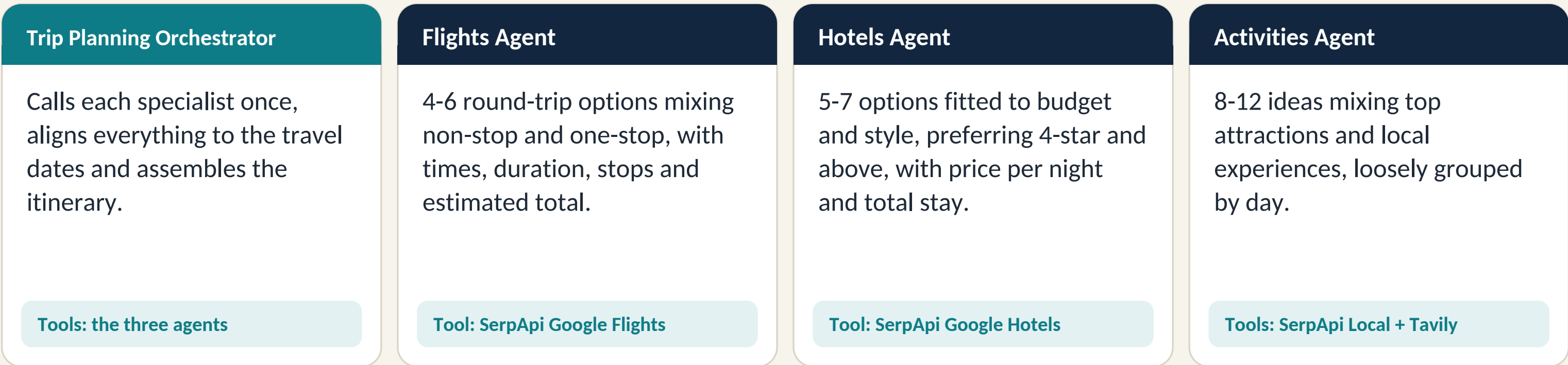
What the router returns

```
{ "intent": "full_plan",  
  "error": true,  
  "error_msg": "To plan your trip I need:  
    departure city, destination city,  
    travel date, number of days",  
  "details": { "source": "", "destination": "",  
    "travel_date": "", "duration_days": 0 } }
```

"A blank field is correct. A guessed field is wrong."

If any of departure city, destination, date or days is missing, the traveller gets a friendly list of exactly what is needed and no search calls are made. For follow-ups the router copies the trip details from the conversation history.

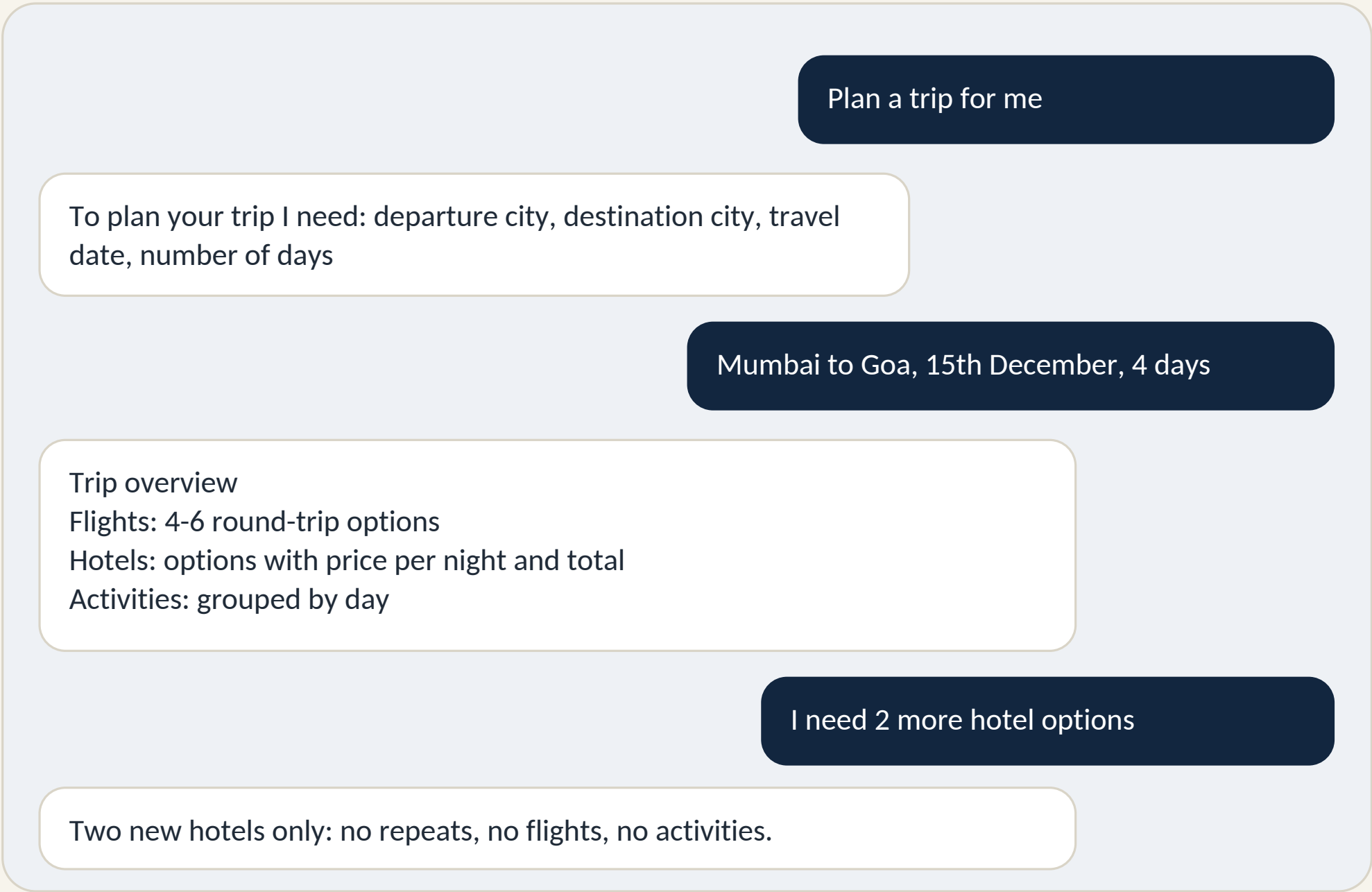
Meet the agents: specialists called once each



Then the Validator steps in

- Travel dates are consistent across flights, hotel and activities
- Activities fit the daily timing
- Hotel nights equal the trip duration
- Small conflicts fixed silently; system details never shown

The experience: three turns, three behaviours



1 error = true

Router flags four missing fields. No search calls are made.

2 full_plan

Orchestrator runs three specialists, then the Validator checks the plan.

3 hotel_only

Memory carries trip details forward; the hotel agent avoids repeats.

Guardrails: trust is the product

Ask, don't guess

Built

Router extracts only what was typed; missing fields trigger a clarifying question.

Ground in tool output

Built

Agents call search tools once and summarise results. Prices are indicative, not bookable.

Validate before replying

Built

A dedicated Validator checks date consistency, stay length and timing.

Fail gracefully

Partial

Retries and friendly messages exist, but one path still ends silently and search failures are unhandled.

Security and dates:

Move API keys to the credential store and rotate the exposed one; add a deterministic date check so no search runs on a past date.

Measuring success

80%
rated acceptable

Human evaluation on 50 test queries (40 of 50). Meets the 80% target.

Reported pilot result

Metric	Target	Status
Plan acceptability	80% or higher	Reported: 80%
Time to first usable plan	50% faster than manual	Baseline to time
Missing-info catch rate	95% or higher	To measure
Follow-up correctness	90% or higher	To measure
Date validity	100%	To measure
Latency p90 (full plan)	Within 60 seconds	To measure

Next cycle: build the 50-query golden set by category (complete, incomplete, follow-up, ambiguous dates, adversarial) and re-run it on every prompt change.

Design review: what we found and fix first

Six of twelve findings shown; the full list is in the specification, section 13

ID	Sev.	Finding	Fix
D-01	P0	API key sits in plain text in an exported workflow node	Rotate; move to credentials
D-02	P1	Web-search query is static placeholder text, not the destination	Bind query to destination
D-03	P1	activity_only intent has no route: the request ends silently	Add Activities branch + fallback
D-04	P1	Router error path can dead-end without a reply	Explicit error branch and message
D-05	P1	Dates stay free text; return date left to the model	Deterministic ISO date node
D-06	P1	Flight Agent input does not receive trip details reliably	Pass router details + message

The point: we ship the review with the product. Known gaps become the roadmap, not surprises.

Roadmap and next steps

NOW

Stabilise (v1.1, 0-2 weeks)

- Fix D-01 to D-06
- Make the Intent Router the production entry
- Write the 50-query golden set
- Time the manual baseline

NEXT

Enrich (v1.2, 2-6 weeks)

- Capture travellers, budget, hotel class, currency
- Deterministic date normalisation
- Wire the activity route
- Day-by-day builder, tracing and evals

LATER

Extend (v2)

- Booking deep links
- Budget optimiser
- Weather and visa checks
- Multi-city, saved preferences, Slack or WhatsApp

Next steps

- Rotate the exposed key today
- Publish results on the golden set
- Ship the six P0/P1 fixes
- Decide: activate v2 as the production entry